

Biometric Integration — Attendance Scenario Guide

How punches become Employee Checkins and Attendance in ERPNext

1) Data flow (high level)

Device punch → sync into ERPNext (pull or push) → process punches into one IN and one OUT per day → ERPNext Auto Attendance generates Attendance → app overrides add deviation fields and optional approval/submission behavior.

2) Required configuration checklist

Employee: set `attendance_device_id` and ensure a Shift Type is assigned.

Shift Type: enable **Auto Attendance** and configure start/end/grace/thresholds.

Biometric Device: set IP/port and sync logs (pull model).

3) Scenarios

A — Normal day (many punches): first punch = IN, last punch = OUT; middle punches ignored by this app's pairing logic.

B — Only 1 punch (missing OUT): IN is created, OUT is missing; auto attendance may not compute working hours. Fix by adding the missing OUT or re-syncing.

C — Late/early entry: the app stores deviation seconds in custom fields (rounded to the nearest minute).

D — Early exit vs shift end: if workflow is Approved, the app can adjust OUT time up to shift end for payroll consistency.

E — Full-time OUT: if $OUT \geq \text{shift end}$, the app can auto-approve and submit Attendance.

F — Overnight shift: if shift end is after midnight, deviation calculations treat end time as next day.

G — Employee not mapped: unknown device IDs are skipped; set Employee `attendance_device_id` and re-sync.

4) Troubleshooting quick checks

- *No Attendance: verify Shift Type Auto Attendance and scheduler.*
- *Wrong times: verify device timezone/clock and timestamp formats.*
- *OUT not updating: app updates OUT only when a later punch exists for that date.*
- *Approval behavior: verify Attendance events in `biometric_integration/hooks.py` and workflow labels.*

This guide describes current app logic; final Attendance results depend on your Shift Type and HR settings.